Institute of Management Sciences
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A Semester Project Report Submitted in Partial Fulfillment
of the Requirements for the Course

Professional Practices

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ABSTRACT

This project looks into the professional side of software engineering things like ethics, accreditation, licensing and the role of professional bodies. I researched frameworks like the ACM/IEEE-CS Software Engineering Code of Ethics, the Ford and Gibbs model of how professions mature and the British Computer Society's Code of Conduct. The main goal was to understand whether software engineering qualifies as a true profession and what responsibilities come with that. I also looked at how these global standards compare to what we have in Pakistan, like the Pakistan Engineering Council (PEC), to make the discussion more relevant to our local context.

INTRODUCTION

Software engineering has become one of the most important fields in the world today. Almost everything around us hospitals, banks, traffic systems, government services runs on software but unlike doctors or lawyers or civil engineers, software engineers do not have the same kind of strict licensing or formal rules governing their work. This raises an important question: is software engineering actually a profession in the full sense of the word or is it still getting there?

This project tries to answer that question by looking at several related topics. I explored what makes a profession "mature" using the Ford and Gibbs framework, what ABET does for educational quality and what organizations like the ACM and IEEE Computer Society actually provide to their members. I also studied the SE Code of Ethics, which was developed jointly by the ACM and IEEE-CS and outlines what software engineers owe to the public, their clients and their colleagues.

The report also considers the British Computer Society's Code of Conduct, the ongoing debate around licensing software engineers and the growing environmental responsibilities of the IT industry. Throughout, references are made to Pakistan's own professional engineering ecosystem including the Pakistan Engineering Council to ensure that these international frameworks are understood within a locally relevant context. Taken together, these topics provide a comprehensive picture of what it means to practise software engineering as a responsible, ethical and accountable profession.

Question 1:

Vision of ABET

ABET stands for the Accreditation Board for Engineering and Technology.

In simpler terms, ABET works to make sure that technical degree programs including computer science and software engineering actually prepare graduates to work in the real world and meet industry and professional standards. They do this by reviewing curricula, faculty, facilities and outcomes against set criteria.

From a Pakistani perspective, institutions like NUST, FAST and COMSATS seek accreditation from bodies like **PEC (Pakistan Engineering Council),** which operates on similar principles to ABET making sure graduates are actually job ready and professionally competent. The idea is the same: ***‘’a degree should mean something”.***

Question 2:

Definition of the Term "Profession"

A profession is an occupation that requires expert knowledge, special technical skills and training which is governed by ethical standards and a commitment to serve society. What separates a profession from just any job is that people in a profession have autonomy, meaning they can work independently, decide their own training, control who can enter the field and obey codes of conduct. The profession is governed internally by its own practitioners and the practitioners are expected to maintain standards not just for their own benefit but for the good of the public.

Examples include medicine, law and engineering. These fields have established educational requirements, licensing or certification systems, governing bodies and ethical codes that members are obligated to follow. Software engineering is still evolving toward full professional status in this sense which is actually one of the core debates this course explores.

Think of it this way: anyone can call themselves a "developer" and build apps but calling yourself a "professional software engineer" ideally implies a certain level of accountability, ethics and formal training. That distinction matters a lot in how the field is regulated and how clients or employers place trust in practitioners.

Question 3:

Ford and Gibbs' Eight Infrastructure Components for Profession Maturity

In their 1996 report A Mature Profession of Software Engineering, Ford and Gibbs outlined eight infrastructure components that define how mature a profession is. These are:

- Initial professional education — a standardized university level curriculum
- Accreditation — formal review and approval of educational programs
- Skills development — ongoing learning and practical training after graduation
- Certification — granted by a professional organization or industry body
- Licensing — legal authorization to practice, usually by a government body
- Professional societies — organizations that represent and support practitioners
- Code of ethics — documented ethical guidelines practitioners are expected to follow
- Professional Development — a well defined and agreed-upon set of knowledge areas that define the field (e.g., SWEBOK — the Software Engineering Body of Knowledge)

These components collectively ensure that a profession maintains quality, accountability and public trust.

Question 4:

Accreditation, Certification and Licensing

Accreditation

Accreditation applies to institutions or programs, not individuals. It refers to the formal evaluation and recognition of educational programs to ensure they meet defined quality standards.

Example: When ABET accredits a Computer Science program at a university, it means that program meets the required academic and professional standards for the field.

Certification

Certification applies to individuals and is typically granted by a professional organization or industry body. It shows that a person has demonstrated a specific level of knowledge or skill usually through an exam or assessment.

Example: The AWS Certified Solutions Architect certification shows that a professional understands cloud architecture at a professional level.

In Pakistan, a developer earning a Google Associate Cloud Engineer cert would be certified in that domain.

Licensing

Licensing is also for individuals but it is backed by law. You cannot legally practice a licensed profession without the required license and violating this has legal consequences.

Example: In many countries, a civil or electrical engineer must be a registered Professional Engineer (PE) to sign off on public infrastructure projects.

In Pakistan, the PEC (Pakistan Engineering Council) issues such licenses for engineers.

To summarize: accreditation is for programs/institutions, certification is voluntary recognition of individual competency and licensing is legally required permission to practice.

Question 5:

Engineering Profession with Highest PE Registration Rate

Among all engineering disciplines, civil engineering is the field where the largest share of working engineers are licensed as Professional Engineers (PEs).This is because civil engineering projects directly affect public safety, such as bridges, buildings and roads, making licensing more critical and often legally required.

In contrast, software engineering has one of the lowest PE registration rates. This is partly because the software industry moves fast, licensing frameworks lag behind and historically the consequences of software failures were less immediately visible though that is changing rapidly as software powers systems like hospitals, banks and transportation.

Question 6:

Advantages and Disadvantages of Licensing Software Engineers

Advantages:

- Public protection — licensing ensures that engineers who work on systems (medical software, banking platforms, air traffic control) meet minimum competency standards
- Accountability — licensed engineers can be held legally responsible for failures, which encourages more careful and ethical work
- Professional credibility — it raises the overall status of software engineering as a profession, similar to medicine or law
- Quality standards — it provides a mechanism for weeding out poorly trained practitioners in high stakes domains

Disadvantages:

- The field evolves too fast — software technologies change so rapidly that any licensing exam risks becoming outdated almost immediately
- Restricts innovation — strict licensing could slow down the entrepreneurial culture that drives the tech industry; many great developers are self-taught
- Defining competency is hard — unlike civil engineering, there is no single agreed upon body of knowledge for software engineering
- Global inconsistency — software is built and deployed globally; licensing in one country means little when teams are spread across continents

In a country like Pakistan where the tech industry is growing rapidly and largely informal, licensing could bring more structure but might also create unnecessary barriers to entry for talented self taught developers.

Question 7:

Software Engineering Code of Ethics and Professional Practices

a. " You must not knowingly use software that has been obtained or kept in ways that are illegal or unethical"

This standard falls under **Principle 4: Judgment.** This principle means being honest and independent when making decisions, including choosing the right tools, software and resources in an ethical way.

b. " Encourage and advance public understanding of software engineering"

This standard falls under **Principle 8: Society** (also referred to in some versions as the "Self" principle). It relates to the responsibility software engineers have to promote public understanding of the discipline and contribute positively to society's awareness of software engineering's impact.

Question 8:

Four ACM Special Interest Groups (SIGs)

The ACM (Association for Computing Machinery) has four Special Interest Groups that are:

- SIGSOFT — Special Interest Group on Software Engineering
- SIGGRAPH — Special Interest Group on Computer Graphics and Interactive Techniques
- SIGAI — Special Interest Group on Artificial Intelligence
- SIGCHI — Special Interest Group on Computer Human Interaction

Question 9:

Journals in the ACM Digital Library

As of the most recent available data, the ACM Digital Library hosts over 50 peer reviewed journals. It covers a wide range of fields, including software engineering, artificial intelligence, data science and human computer interaction.

The exact count changes periodically as new publications are added to the library.

Question 10:

Four IEEE Computer Society Technical Committees

The IEEE Computer Society includes:

- Technical Committee on Software Engineering (TCSE)
- Technical Committee on Computer Architecture (TCCA)
- Technical Council on Data Engineering
- Technical Committee on Security and Privacy (TCSP)

Question 11:

Transactions Publications in the IEEE-CS Digital Library

It includes more than 20 different journals:

- IEEE Transactions on Software Engineering
- IEEE Transactions on Computers
- IEEE Transactions on Dependable and Secure Computing
- IEEE Transactions on Neural Networks and Learning Systems

The exact number changes over time because new journals are added and some are discontinued.

Question 12:

Moral Justifications for Illegal Computer Access and Counter-Arguments

Justification 1: Curiosity or Learning

An individual may argue that accessing a system illegally helps them learn and improve technical skills.

Counterargument:

Learning does not justify violating laws or privacy. Ethical learning should occur through legal platforms such as labs, simulations or authorized environments.

Justification 2: Exposing Security Weaknesses

A hacker may claim they are identifying vulnerabilities for the public good.

Counterargument:

Security testing must be authorized. Unauthorized access can cause harm and legal consequences, even if intentions are good.

Question 13:

British Computer Society (BCS) Code of Conduct

The British Computer Society's Code of Conduct organizes its professional standards into four main groups. Here are three of them with examples:

1. Public Interest

This group of standards requires IT professionals to act in the interest of the general public, not just their employers or clients. Practitioners must consider the social, legal and ethical implications of their work on the broader public.

Example standard: **"You must give proper consideration to public health, privacy, security and the wellbeing of others, as well as the protection of the environment"**

This means, for instance, that a software developer should not build a system that collects user data in a way that violates privacy rights, even if instructed to do so by an employer.

2. Professional Competence and Integrity

This group ensures that IT professionals only take on work they are genuinely qualified to do and that they continue to develop their skills throughout their careers.

Example standard: **"You should agree to perform work or provide services only when they fall within your professional expertise"**

A developer who has no experience with safety critical embedded systems, for example, should not take on a contract to build software for medical devices without appropriate supervision or training.

3. Duty to Relevant Authority

This group governs the professional's responsibilities toward their employer, client or other authority. It requires acting within the law, carrying out duties with loyalty and not acting against the legitimate interests of the authority while also maintaining professional ethics.

Example standard: **"You must not provide false, misleading, or incomplete information about the performance of products, systems, or services"**

This means an IT professional must honestly report to management if a project is behind schedule or a system has known flaws, rather than hiding the truth to avoid uncomfortable conversations.

Question 14:

Environmental Impact of the IT Industry and the Role of IT Professionals
The Problem: Data Centers and Energy Consumption

The IT industry has a significant and growing environmental footprint. One of the biggest contributors is data centers , the server farms that power everything from cloud storage to social media platforms and AI models. These facilities consume enormous amounts of electricity to run servers and keep them cool. Globally, data centers are estimated to account for around 1-2% of worldwide electricity consumption and that share is growing rapidly as demand for AI computing, streaming and cloud services surges.

Beyond data centers, the manufacturing of hardware laptops, phones, chips involves rare earth minerals and generates significant e-waste.

In Pakistan, where e-waste disposal infrastructure is limited, discarded electronics often end up in informal recycling operations that release toxic substances into the environment.

What IT Professionals Can Do

IT professionals are positioned to be part of the solution. Here are some practical ways to contribute:

- Write efficient code — inefficient software wastes computational resources. Optimizing algorithms and reducing unnecessary computations directly reduces energy use
- Advocate for green hosting — choose or recommend cloud providers that run on renewable energy, such as Google Cloud or Microsoft Azure's green infrastructure
- Promote responsible hardware use — support policies that extend device lifespans, enable repairability and ensure proper e-waste recycling at the organizational level
- Design for sustainability — when building software systems, consider the computational cost of features and architecture decisions

In a country like Pakistan, where energy shortages are already a major challenge, writing efficient software is not just an environmental virtue it is a practical necessity. Every unnecessary computation cycle has a real cost in electricity, which has real costs on the national grid and carbon emissions. IT professionals here have an especially compelling reason to take this seriously.

CONCLUSION

Working through this project taught me a lot about what it means to be a professional not just technically, but ethically and socially too.

First, software engineering is still maturing as a profession. It has made real progress there is a body of knowledge, there are professional societies and there is a code of ethics but it has not reached the same level of formal recognition as fields like medicine or civil engineering. The debate around licensing shows this clearly; people in the industry genuinely disagree about whether it would help or hurt.

Second, the ACM/IEEE-CS Code and the BCS Code of Conduct both make clear that software engineers have real obligations to users, to the public and to society as a whole. Given how much software affects people's lives now, that responsibility is serious.

Third, environmental impact is something the IT industry can no longer ignore. Things like writing efficient code or pushing for green hosting might seem small but they add up. In Pakistan especially, where the power grid is already under pressure, these choices have a direct real world impact.

The biggest takeaway for me is that being a software engineer is about more than writing code. It comes with accountability, ethical judgment, and a responsibility to the people your work affects.